

Minimum Viable Product Plan Vision & Justifications

Product Vision

For U.S. residents and public-health organizations responding to a pandemic, the CDC Outbreak Response Platform provides timely, privacy-conscious reporting and regional visibility into cases, deaths, testing activity, and test-kit supply, enabling officials to direct funding and kits where they are needed most and reduce further spread.

Product Roadmap Justifications

Phase	Feature	Justification
1	Basic Dashboard with heatmap display	A basic, non-interactive web heatmap directly satisfies a critical first-release outcome while requiring fewer dependencies than mobile, customizable, or automated alternatives. It provides immediate outbreak visibility after the foundational database, portal, authentication, and manual data intake are available.
2	Security Update	Security is strengthened after the initial role and data flows are established and before broader exports, searches, and regional communications increase exposure. This sequence protects the growing product without delaying the simplest usable outbreak-reporting capability.

3	Automated Integration for Importing	Automation follows manual importing so data formats, validation rules, and operating workflows can stabilize first. It then reduces reporting delay and manual workload as the number of participating organizations grows.
4	In-App messaging for Doctors and Patients	In-app messaging is valuable but non-critical to the stated first-release outcomes and depends on mature accounts, mobile experiences, security controls, and communication governance. It is therefore deferred until the core reporting and decision-support functions are reliable.

MVP User Story Justifications

Included in MVP	User Story	Justification
Yes	As a Patient, I can report a positive virus test and my U.S. location so that the CDC can track the spread of the outbreak.	This story is indispensable because universal U.S. case reporting is an explicitly critical outcome and supplies the location-based data required by the outbreak heatmap and regional prioritization decisions. Scheduling it in Sprint 1 establishes the source data needed for the Sprint 2 heatmap and the Sprint 3 allocation analysis; without it, the MVP cannot observe spread or direct the response reliably.

No	As a Test Center, I can receive a low-stock warning when inventory crosses a defined threshold so that I can request additional supplies before testing is interrupted.	The warning is valuable, but it is an automated convenience rather than the minimum capability needed to track testing and supply. In the MVP, test centers can report tests administered and the CDC can compare activity with reported production. Its Sprint 5 placement deliberately follows the Sprint 4 inventory workflow, allowing threshold automation to be built only after inventory behavior and data quality have stabilized.
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Sprint Scenario Responses

Sprint	Scenario	Response
1	One user story depends on another system that will not be available until the next sprint.	The team should first identify which of the three Sprint 1 stories is blocked and confirm that the dependency cannot be separated from its acceptance criteria. The Product Owner and developers should then remove that story from the sprint forecast, return it to the ordered backlog, document and coordinate the external dependency, and schedule it no earlier than Sprint 2. If capacity remains and the Sprint 1 goal is preserved, the team can pull the highest-priority dependency-free story of comparable size into the sprint.

3	A software engineer is sick and will miss most of the sprint.	The team should reforecast capacity immediately against the three Sprint 3 stories: CDC allocation ranking, the patient heatmap and guidance view, and the doctor treatment submission. The developers should swarm or reassign work that can be completed safely, preserve the Sprint Goal by completing the highest-value feasible work first, and defer the lowest-value unfinished scope if necessary. The Product Owner should reorder any removed work, communicate any threat to the MVP forecast, and avoid compensating through unsustainable overtime or reduced quality.
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